Time to sort 1 string → $O(m \log m)$

Time to sort n strings → $O(nm \log m)$. — 1

Time to sort a list of ints → $O(n \log n)$.

Time ~~to take~~ to compare 2 ints → $O(1)$

No. of comparisons in sorting → $O(n \log n)$

Time to compare 2 strings → $O(m)$

Time to sort a list of strings → $O(nm \log n)$. — 2

Overall, $TC = O(nm \log m + mn \log n)$

$= O(nm(\log n + \log m))$.